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MID-TERM EXAMINATION

Course: STAT452	
Time: 90 minutes	Term: 3 – Academic year: 2023-2024
Lecturer(s): **Dr Nguyễn Thị Mộng Ngọc**	
Student name:	Student ID:

(Notes: Neither books nor laptops, phones allowed)

Q.1 (Efficiency of computer programs)

A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of processed requests per hour. Applying the program to data sets of different sizes, she gets the following results,

Data size (gigabytes), x	5	6	6	7	7	8	8	9	10	10	11	12	13	14	14	15	16
Processed requests, y	43	40	55	50	41	39	23	17	22	26	21	19	20	21	23	14	16

In general, larger data sets require more computer time, and therefore, fewer requests are processed within 1 hour. The response variable here is the number of processed requests (y), and we attempt to predict it from the size of a data set (x).

The following R output is from a simple linear regression of y on x:

```
Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max
-15.0222  -3.9593   0.0616   4.0826  13.9150

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  59.2107     5.9826   9.897 5.72e-08 ***
x            -3.0209     0.5643  -5.354 8.04e-05 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.797 on 15 degrees of freedom
Multiple R-squared:  0.6565,Adjusted R-squared:  0.6336
F-statistic: 28.66 on 1 and 15 DF,  p-value: 8.041e-05
```

Use the data and the R output above to answer the following questions.

(a) Obtain the equation of the least squares line and interpret its slope.

(b) Use the output to calculate a confidence interval with a confidence level of 95% for the slope of the population regression line, and interpret the resulting interval.

(c) What proportion of observed variation in 'Processed requests'(the number of processed requests) can be attributed to the approximate linear relationship between 'Processed requests' and 'Data size'(the size of a data set)?

(d) Does there appear to be a useful linear relationship between 'Processed requests' and 'Data size'? State and test the appropriate hypotheses using a significance level of 5%.

(e) What is the value of the sample correlation coefficient?

(f) What value of 'Processed requests' would you predict when 'Data size' is 15 gigabytes, and what is the value of the corresponding residual?

Q.2 (Modeling the Energy Content of Municipal Solid Waste)

Efficient design of certain types of municipal waste incinerators requires that information about energy content of the waste be available. The authors of the article *"Modeling the Energy Content of Municipal Solid Waste Using Multiple Regression Analysis" (J. of the Air and Waste Mgmnt. Assoc., 1996: 650–656)* kindly provided us with the accompanying data on y = energy content (kcal/kg), the three physical composition variables x_1 = % plastics by weight, x_2 = % paper by weight, and x_3 = % garbage by weight, and the proximate analysis variable x_4 = % moisture by weight for waste specimens obtained from a certain region.

The following R output (missing) is from a multiple linear regression of y on (x_1, x_2, x_3, x_4):

```
> summary(mod2)

Call:
lm(formula = Energy.content ~ ., data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-41.34  -24.04  -11.00   22.54   59.73

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2245.093    177.892  12.621 2.42e-12 ***
Plastics       28.922      2.823  10.243 1.97e-10 ***
Paper           7.643      2.314   3.303  0.00288 **
Garbage         4.297      1.916   2.243  0.03404 *
Water         -37.356      1.834 -20.367  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.48 on 25 degrees of freedom
Multiple R-squared:  0.9641,Adjusted R-squared:  0.9583
F-statistic: 167.7 on 4 and 25 DF,  p-value: < 2.2e-16

> simpleAnova(mod2)
Analysis of Variance Table

Response: Energy.content
           Df Sum Sq   Mean Sq   F value    Pr(>F)
Predictors  ?    ?      166234       ?     < 2.2e-16 ***
Residuals  25  24775        ?
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Use the R output above to answer the following questions.

(a) Complete the Analysis of Variance Table.

(b) What are the values of SST and the coefficient of determination? And interpret the resulting coefficient .

(c) State and test the appropriate hypotheses to decide whether the model fit to the data specifies a useful linear relationship between energy content and at least one of the four predictors.

(d) What is a point estimate of the error standard deviation σ, and how would you interpret it?

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